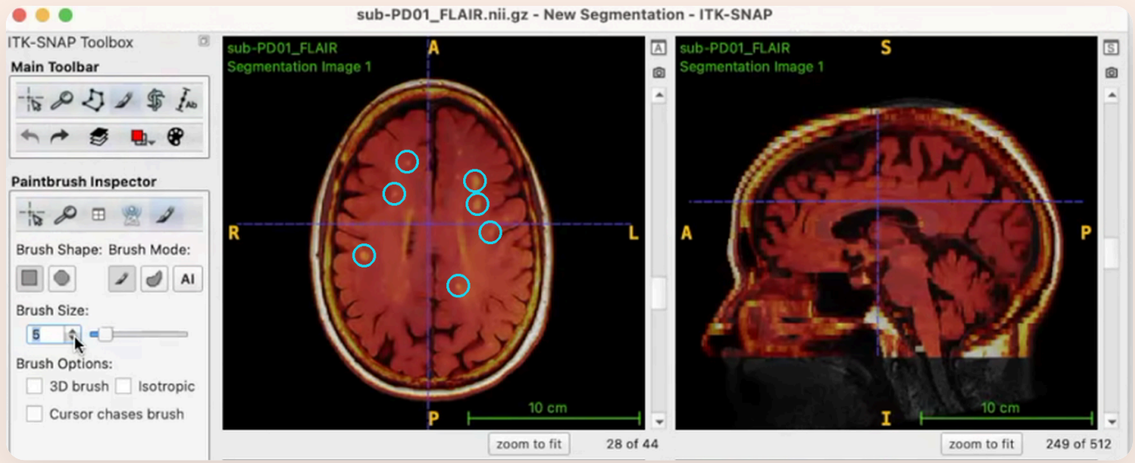
LESSON 5 · CAPSTONE OUTPUT

From Manual Segmentation to a Group Comparison

One student's run of the capstone on the six-participant teaching dataset

1 Manual segmentation of FLAIR hyperintensities

ITK-SNAP paintbrush, label 1, one mask per participant. Axial FLAIR of sub-PD01; circles (added for this figure) mark example lesion-like hyperintensities.



2 Marked-voxel count per mask

measure_lesion_masks.sh counts label-1 voxels in each saved mask. Both controls at 0 are saved masks with nothing marked.

```
Measuring masks saved from ITK-SNAP...
Measured sub-PD01: 6528 label-1 voxels
Measured sub-PD02: 5645 label-1 voxels
Measured sub-PD03: 2142 label-1 voxels
Measured sub-CON01: 0 label-1 voxels
Measured sub-CON02: 0 label-1 voxels
Measured sub-CON03: 464 label-1 voxels
```

3 Group comparison

Parkinson's vs. control, n = 3 each. Dots = participants, bar = group mean, whisker = ±1 sample SD (lower end truncated at 0). Counts are voxels, not volume. Plotted with plot_lesion_groups.m (MATLAB/Octave).

